

A NINE-VERTEX P_4 -SPARSE MINIMAL $(2, 3)$ -POLAR OBSTRUCTION THAT IS NOT A COGRAPH

ABSTRACT. Contreras-Mendoza and Hernández-Cruz asked, as Problem 1 of their paper on minimal obstructions for polarity in generalizations of cographs, whether every P_4 -sparse minimal (s, k) -polar obstruction is a cograph. We answer this in the negative at $(s, k) = (2, 3)$. The witness is the 9-vertex, 15-edge thin spider H^* with head $2K_2 + K_1$: it is P_4 -sparse, its unique induced P_4 shows that it is not a cograph, it is not $(2, 3)$ -polar, and all nine of its one-vertex deletions are. Completion gives a second counterexample at $(s, k) = (3, 2)$. The graph is exhibited by its edge list, and the proof is a short case analysis together with nine explicit polar partitions, so no computation is needed to check it.

1. INTRODUCTION

All graphs are finite, simple and undirected. A k -cluster is a disjoint union of at most k cliques. For positive integers s and k , a graph G is (s, k) -polar if $V(G)$ has a partition (A, B) such that $G[A]$ is a complete multipartite graph with at most s parts and $G[B]$ is a k -cluster; such a pair (A, B) is an (s, k) -polar partition. Either side may be empty. Since $(1, 1)$ -polar is exactly split, polarity generalises the Földes–Hammer notion [4]. The property is hereditary, so it is characterised by its *minimal obstructions*: graphs that are not (s, k) -polar although every proper induced subgraph is. Consequently a graph is a minimal (s, k) -polar obstruction as soon as it is not (s, k) -polar and all of its one-vertex deletions are.

A *cograph* is a P_4 -free graph, and a graph is P_4 -sparse if no five of its vertices induce more than one P_4 ; every cograph is P_4 -sparse. Contreras-Mendoza and Hernández-Cruz determined the minimal (s, k) -polar obstructions in these classes when $\min\{s, k\} \leq 1$ and posed the following question [1, Problem 1].

Problem 1. For any positive integers s and k , is every P_4 -sparse minimal (s, k) -polar obstruction a cograph?

The problem carries this number both in the version of record of [1] and in version 2 of its arXiv e-print, and it is quoted verbatim above.

Two further facts from [1] locate the question. First, a P_4 -sparse minimal (s, k) -polar obstruction has at most $(s + 1)(k + 1)$ vertices [1, Theorem 57], a

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bound the authors credit to Hannebauer’s thesis [5]. Second, the authors’ sequel proves that there are exactly 50 P_4 -sparse minimal 2-polar obstructions and that each of them is a cograph [2]; here 2-polar means $(2, 2)$ -polar, and the minimal 2-polar obstructions inside the smaller class of cographs had been determined earlier [6]. Together with the settled range $\min\{s, k\} \leq 1$, this leaves $(s, k) = (2, 3)$, where the bound reads 12, as the smallest cell not covered by those results. That is the cell we settle.

Theorem 1. *Let H^* be the graph on $V = \{s_1, s_2, k_1, k_2, a_1, a_2, b_1, b_2, x\}$ whose edge set consists of exactly the following fifteen pairs and no others:*

$$\begin{aligned} & s_1k_1, \quad s_2k_2, \quad k_1k_2, \\ & k_1a_1, \quad k_1a_2, \quad k_1b_1, \quad k_1b_2, \quad k_1x, \\ & k_2a_1, \quad k_2a_2, \quad k_2b_1, \quad k_2b_2, \quad k_2x, \\ & a_1a_2, \quad b_1b_2. \end{aligned}$$

Then H^ is P_4 -sparse, it is not a cograph, and it is a minimal $(2, 3)$ -polar obstruction. In particular, the answer to Problem 1 is no.*

Writing $S = \{s_1, s_2\}$, $K = \{k_1, k_2\}$ and $R = \{a_1, a_2, b_1, b_2, x\}$, the graph H^* is the spider [7] with independent set S , clique K , the two legs s_1k_1 and s_2k_2 , and head R , where K is complete and S is anticomplete to R and $H^*[R] = 2K_2 + K_1$. Its degree sequence is $(1, 1, 7, 7, 3, 3, 3, 3, 2)$, summing to $30 = 2 \cdot 15$, and the remaining 21 of the $\binom{9}{2} = 36$ pairs are non-edges. In `graph6` format, with the vertices in the order $s_1, s_2, k_1, k_2, a_1, a_2, b_1, b_2, x$, the graph is the seven-character string

`HRKx`co`

and the same order presents the fifteen edges above as the 0-indexed list

02 13 23 24 25 26 27 28 34 35 36 37 38 45 67.

The graph is also $2K_2$ -split: V is the disjoint union of the clique K , the independent set $\{s_1, s_2, x\}$, and the set $\{a_1, a_2, b_1, b_2\}$, which induces $2K_2$.

Corollary 2. *The complement $\overline{H^*}$ is a P_4 -sparse minimal $(3, 2)$ -polar obstruction that is not a cograph. Hence the answer to Problem 1 is also no at $(s, k) = (3, 2)$.*

Proof. A graph G is (s, k) -polar if and only if $\overline{G}$ is (k, s) -polar: complementation exchanges “complete multipartite with at most s parts” and “disjoint union of at most s cliques”, and it commutes with taking induced subgraphs, so it also exchanges minimal (s, k) -polar obstructions with minimal (k, s) -polar obstructions. Complementation fixes the set of induced P_4 ’s of a graph, because $\overline{P_4} \cong P_4$, so it preserves both P_4 -sparseness and the property of not being a cograph. Now apply Theorem 1. $\square$

Remark. What is refuted is Problem 1 as a statement quantified over all pairs (s, k) . The settled range $\min\{s, k\} \leq 1$ and the case $(2, 2)$ are theorems of [1] and [2] and remain so, and nothing here bears on any other cell with $\min\{s, k\} \geq 2$, or on the classification, or even the finiteness, of the P_4 -sparse minimal $(2, 3)$ -polar obstructions; in particular we do not claim that

9 is the least order of a non-cograph example. Finally, H^* is not among the 50 obstructions of [2], since $H^* - x$ is not (2,2)-polar and therefore H^* is not a minimal (2,2)-polar obstruction at all.

2. PROOF OF THEOREM 1

Throughout, S , K and R are as above. We use two elementary observations repeatedly: every vertex of K is adjacent to every vertex of R and to the other vertex of K ; and s_i has degree 1, its unique neighbour being k_i .

Step 1: the only induced P_4 of H^ is $s_1k_1k_2s_2$; hence H^* is P_4 -sparse and is not a cograph.* The set $K \cup R$ induces the join of K with $H^*[R] = 2K_2 + K_1$; a join of two cographs is a cograph, so no induced P_4 lies inside $K \cup R$ and every induced P_4 of H^* meets S . Suppose Q induces a P_4 and $s_1 \in Q$. As s_1 has degree 1, it is an endpoint of that path and $k_1 \in Q$. The third vertex of the path is a neighbour of k_1 other than s_1 , and the fourth is adjacent to the third but not to k_1 ; the only vertex of H^* other than s_1 that is not adjacent to k_1 is s_2 , so the fourth vertex is s_2 and the third is its unique neighbour k_2 . Thus $Q = \{s_1, k_1, k_2, s_2\}$, and symmetrically if $s_2 \in Q$. Conversely $s_1k_1k_2s_2$ is an induced path, since s_1s_2 , s_1k_2 and s_2k_1 are non-edges. So H^* has exactly one induced P_4 : no five vertices can contain two of them, so H^* is P_4 -sparse, and H^* is not P_4 -free, so it is not a cograph.

For the remaining two steps, note that a complete multipartite graph with at most two parts is exactly a complete bipartite graph $K_{p,q}$ with $p+q \geq 1$, $p, q \geq 0$; in particular it is triangle-free. We record the only property of it we need.

Lemma 3. *Let $G[A]$ be complete multipartite with at most two parts and let uv be an edge with $u, v \in A$. Then every $w \in A$ is adjacent to u or to v .*

Proof. Parts are independent, so u and v lie in different parts, and there are at most two parts, so w lies in the part of u or in the part of v . Distinct parts are completely joined, so in the first case w is adjacent to v and in the second to u . $\square$

Lemma 4. *Let (A, B) be a partition of $V(H^*)$ with $H^*[B]$ a 3-cluster. If $B \cap K \neq \emptyset$, then $B \cap R$ is a clique, and consequently $\{a_1, a_2\} \subseteq A$ or $\{b_1, b_2\} \subseteq A$.*

Proof. Let $k \in B \cap K$. As k is adjacent to every vertex of R , the set $(B \cap R) \cup \{k\}$ lies in a single connected component of $H^*[B]$, and components of a cluster are cliques; hence $B \cap R$ is a clique. The cliques of $H^*[R] = 2K_2 + K_1$ are the subsets of $\{a_1, a_2\}$, of $\{b_1, b_2\}$ and of $\{x\}$, so $B \cap R$ misses one of the two pairs $\{a_1, a_2\}$, $\{b_1, b_2\}$ entirely, and that pair lies in A . $\square$

Step 2: H^ is not (2,3)-polar.* Suppose (A, B) were a (2,3)-polar partition. We distinguish two cases.

Case $B \cap K \neq \emptyset$. By Lemma 4 the set A contains one of the edges a_1a_2 , b_1b_2 ; call it uv . If the other vertex k' of K lies in A , then k' is adjacent to both u and v , so $\{k', u, v\}$ is a triangle in A , contradicting the fact that $H^*[A]$ is complete bipartite. So $K \subseteq B$. Then $s_1 \in A$: for if $s_1 \in B$, its unique neighbour k_1 lies in B as well, so s_1 and k_1 are in one component of $H^*[B]$, a clique that also contains k_2 because k_1k_2 is an edge, while s_1k_2 is a non-edge — impossible. But s_1 is adjacent to neither u nor v , since S is anticomplete to R , and this contradicts Lemma 3 applied to the edge uv of A .

Case $K \subseteq A$. No vertex $r \in R$ can lie in A , since $\{k_1, k_2, r\}$ would be a triangle; hence $R \subseteq B$. The three components of $H^*[R]$ are pairwise anticomplete and $K \cap B = \emptyset$, so they are already three distinct components of $H^*[B]$. If some s_i lay in B , it would be a fourth component, its unique neighbour k_i being in A ; then $H^*[B]$ would have at least four components and could not be a 3-cluster. So $S \subseteq A$ and $A = \{s_1, s_2, k_1, k_2\}$, which induces the path $s_1k_1k_2s_2$. Apply Lemma 3 to the edge s_1k_1 of A and to $w = s_2$: the vertex s_2 would have to be adjacent to s_1 or to k_1 , and it is adjacent to neither.

Both cases are impossible, so no $(2,3)$ -polar partition of H^* exists.

Step 3: every one-vertex deletion of H^ is $(2,3)$ -polar.* The nine partitions below are $(2,3)$ -polar partitions of the nine graphs $H^* - v$. For each we list A split into its parts, and B split into its clique components, so that each line can be checked against the definition by inspection: the parts are independent sets pairwise completely joined, the components are cliques pairwise anticomplete, and $A \cup B = V \setminus \{v\}$.

v	parts of A	clique components of B
s_1	$\{k_2\} \mid \{k_1, s_2\}$	$\{a_1, a_2\} \mid \{b_1, b_2\} \mid \{x\}$
s_2	$\{k_1\} \mid \{k_2, s_1\}$	$\{a_1, a_2\} \mid \{b_1, b_2\} \mid \{x\}$
k_1	$\{k_2\} \mid \{s_2, x\}$	$\{a_1, a_2\} \mid \{b_1, b_2\} \mid \{s_1\}$
k_2	$\{k_1\} \mid \{s_1, x\}$	$\{a_1, a_2\} \mid \{b_1, b_2\} \mid \{s_2\}$
x	$\{k_1\} \mid \{k_2, s_1\}$	$\{a_1, a_2\} \mid \{b_1, b_2\} \mid \{s_2\}$
a_1	$\{s_1, s_2, a_2, x\}$	$\{k_1, k_2, b_1, b_2\}$
a_2	$\{s_1, s_2, a_1, x\}$	$\{k_1, k_2, b_1, b_2\}$
b_1	$\{s_1, s_2, b_2, x\}$	$\{k_1, k_2, a_1, a_2\}$
b_2	$\{s_1, s_2, b_1, x\}$	$\{k_1, k_2, a_1, a_2\}$

For the first five rows, A has two parts and B is a 3-cluster; for the last four, A is independent (one part) and B is the clique K_4 induced by K together with the surviving pair of the head. Two rows deserve a word. In the row $v = k_1$ the vertex s_1 becomes isolated because its only neighbour has been deleted, which is why it can be a clique component of B on its own; the pair $\{s_2, x\}$ is independent and both of its vertices are adjacent to k_2 , so $H^*[A]$ is $K_{1,2}$. In the row $v = a_1$ the set $\{s_1, s_2, a_2, x\}$ is independent,

because S is anticomplete to R and a_2 's only neighbour inside R was a_1 ; and $\{k_1, k_2, b_1, b_2\}$ is a K_4 since K is a clique complete to R and b_1b_2 is an edge.

By Step 2 and Step 3, H^* is a minimal $(2, 3)$ -polar obstruction, and by Step 1 it is P_4 -sparse and not a cograph. $\square$

3. VERIFICATION

Everything asserted about H^* in Theorem 1 is decided by the hand argument of Section 2 from the edge list printed in Section 1, and a reader who wants to check it needs no computer. The claims were nevertheless re-derived independently, by a program written from the printed strings alone and using exact bitmask and integer arithmetic with no third-party library: it decodes the `graph6` string, checks that the decoded edge set is *equal* to the printed labelled edge list, and then recomputes the degree sequence, the spider structure, the single induced P_4 over all $\binom{9}{4} = 126$ four-subsets, the P_4 -sparseness over all $\binom{9}{5} = 126$ five-subsets, the failure of $(2, 3)$ -polarity by exhausting all $2^9 = 512$ bipartitions, the nine deletion certificates of Step 3 one at a time against the definition, and independently of them a $(2, 3)$ -polar partition of each deletion found by exhaustion; it also recomputes Corollary 2 on the complement. All 71 of its checks passed. The program `verify.py` and the transcript `verify.output.txt` of the run accompany this note; they are not needed in order to check the theorem.

Relation to the literature. The nearest published work is the same authors' characterisation of polarity on H -split graphs [3]: H^* is a $2K_2$ -split graph, it satisfies the order bound $s + 2k + 2 = 10$ proved there, and that characterisation confirms clause by clause both the failure of $(2, 3)$ -polarity and the nine certificates above; it is not applied there to P_4 -sparse graphs or to cographs. We make no priority claim. In particular we have not inspected [5], the 2010 thesis on matrix colorings of P_4 -sparse graphs from which the order bound $(s + 1)(k + 1)$ is quoted in [1].

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